

Utility & Demand

Based on Ch. 8, Economics 10th ed., Parkin, Powell & Mathews

Course: Eco 101 (Introduction to Microeconomics)

Rational Consumer & Utility

- A **rational consumer** is someone who clearly knows what they want and makes the best possible use of available opportunities.
- **Utility** refers to the *satisfaction* or *happiness* a consumer gains from consuming goods or services.
- An individual's consumption bundle is the full collection of goods and services that the person consumes.
- A *utility function* shows the *total utility* (satisfaction) generated from an individual's consumption bundle.

Consumption Possibilities & Budget Constraints

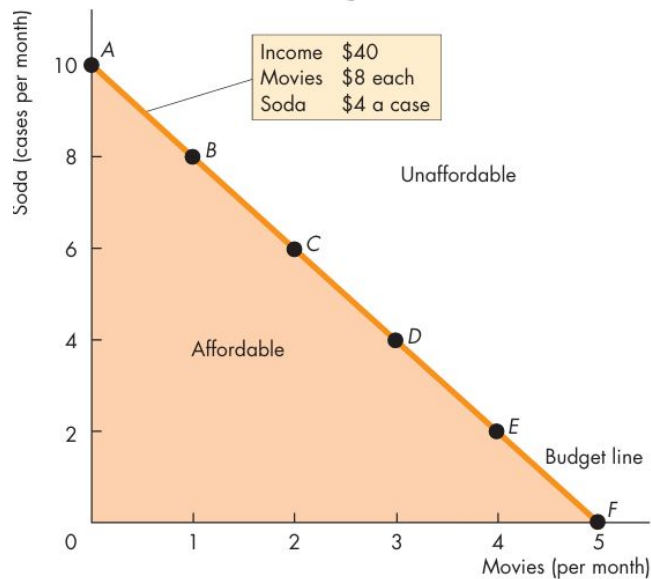
- Consumption possibilities include all combinations of goods and services you can afford.
- What you can buy is limited by your **income** and the **prices** of goods and services.
- To simplify analysis, consumption is often modeled using **two goods**.
- A **budget line** shows the limit of what a consumer can buy when all income is spent.

Affordable vs Unaffordable Bundles, Opportunity Cost & Preferences

- The budget line separates affordable combinations from those that are unaffordable.
- Every bundle on or inside = affordable
- Every bundle outside = unaffordable.
- **Opportunity cost** explains the slope of the graph
- **Preferences** are what a consumer chooses/ prefers..

Figure 8.1 illustrates Lisa's consumption possibilities of movies and soda and her budget line. Lisa has an income of \$40 a month, the price of a movie is \$8, and the price of soda is \$4 a case.

FIGURE 8.1 Lisa's Budget Line



Possibility	Movies		Soda	
	Quantity	Expenditure (dollars)	Cases	Expenditure (dollars)
A	0	0	10	40
B	1	8	8	32
C	2	16	6	24
D	3	24	4	16
E	4	32	2	8
F	5	40	0	0

The graph and the table show six possible ways in which Lisa can allocate \$40 to movies and soda. In row C and at point C, she sees 2 movies and buys 6 cases of soda. The line AF is Lisa's budget line and is a boundary between what she can afford and what she cannot afford. Her choices must lie along the line AF or inside the orange area.

The Consumer's Choice Problem

- Given a \$40 per month budget, which point on the budget line will the consumer choose?
- We want to find the consumption bundle (the point on the budget line) that maximizes our **total utility**.
- **Optimal consumption bundle** = the consumption bundle that maximizes the *total utility* given the budget constraint
- Finding the optimal consumption choice is a “*how much*” problem i.e. we need to determine how much to spend on each good.

Total Utility, Marginal Utility & Diminishing Marginal Utility

Total Utility

- The total benefit that a person gets from the consumption of all the different goods and services is called total utility.
- Total utility depends on the level of consumption—more consumption generally gives more total utility.

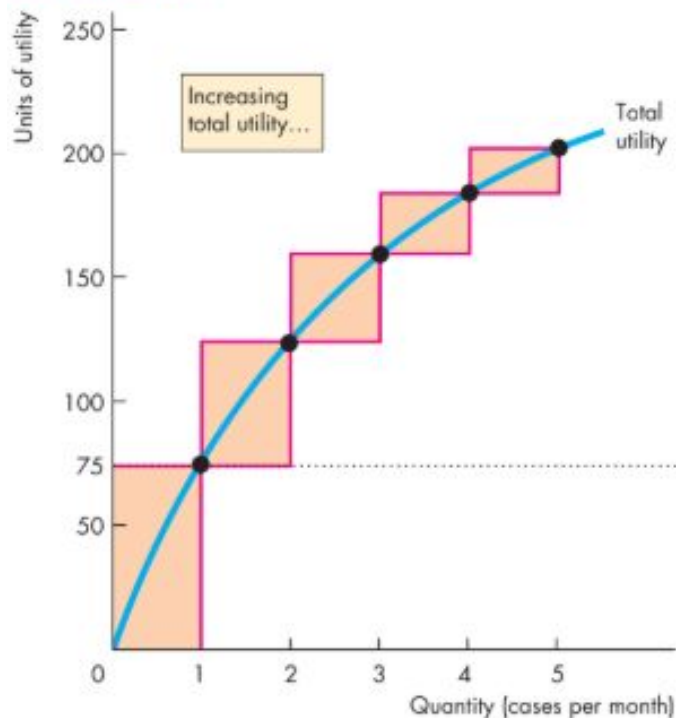
Marginal Utility

- Marginal utility is the change in total utility that results from a one-unit increase in the quantity of a good consumed.

Principle of Diminishing Marginal Utility

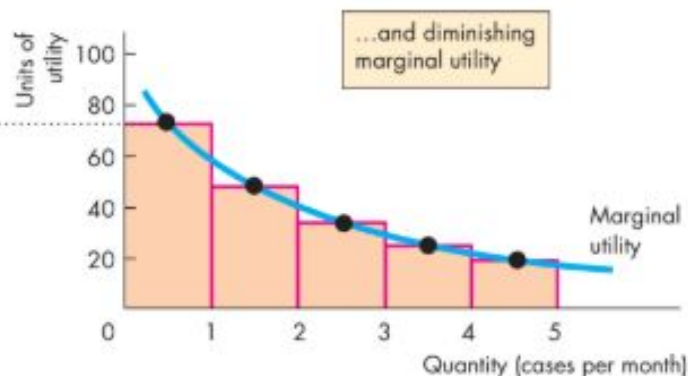
- The tendency for marginal utility to decrease as the consumption of a good increases is due to the principle of diminishing marginal utility.

FIGURE 8.2 Total Utility and Marginal Utility



(a) Total utility

The figure graphs Lisa's total utility and marginal utility from soda based on the numbers for the first 5 cases of soda a month in Table 8.1. Part (a) shows her total utility—increasing total utility. The bars along the total utility curve show the extra total utility from each additional case of soda—marginal utility. Part (b) shows Lisa's diminishing marginal utility from soda.



(b) Marginal utility

Marginal Utility per Dollar & Marginal Decision-Making

- The marginal decision is a question of how to spend the marginal dollar..
- Calculate the marginal utility per dollar spent on either pepsi or pizzas requires determination of how much additional utility you get from spending an additional dollar on either good.
- The marginal utility per dollar spent on a good or service is the additional utility from spending one more dollar on that good or service.

Diminishing MU and Its Effect on MU per Dollar

- Due to **diminishing marginal utility**, marginal utility of a good falls as the quantity of that good consumed rises.--- water, video games, dolls as you grow up....
- Therefore, marginal utility per dollar spent on each good declines as the quantity of that good consumed rises, because of diminishing MU.
- Hence, if your marginal utility per dollar spent on one good is higher than that per dollar spent on the other → you should spend \$1 less on the first good and \$1 more on the other good.

The Equi-marginal Principle

- The equi-marginal principle explains how consumers maximize utility by allocating income so that the marginal utility per dollar is equal across all goods.
- This is the **utility-maximizing principle of marginal analysis** (Equi-Marginal Principle)= Consumer Equilibrium (The marginal utility per dollar spent must be the same for all goods and services in the optimal consumption bundle)
- Generally, **marginal utility per dollar spent on a good = Marginal utility of one unit of the good/Price of one unit of the good.**

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

- That is, at the optimal consumption bundle

Say there is a rational person with a budget of \$10. He/she wants to choose a combination of burgers and juice that gives them the maximum possible utility..

Juice		Burgers	
Glasses (\$2 per glass)	Total Utility (Utils)	No. of burgers (\$1 per burger)	Total Utility (Utils)
1	30	1	13
2	55	2	25
3	75	3	36
4	90	4	46

From the given information, find out how many burgers and glasses of juice will the consumer choose?